Business Intelligence Systems

INFO 4555/5556

Fall 2025

WhereScape RED



Cornell University

WhereScape RED

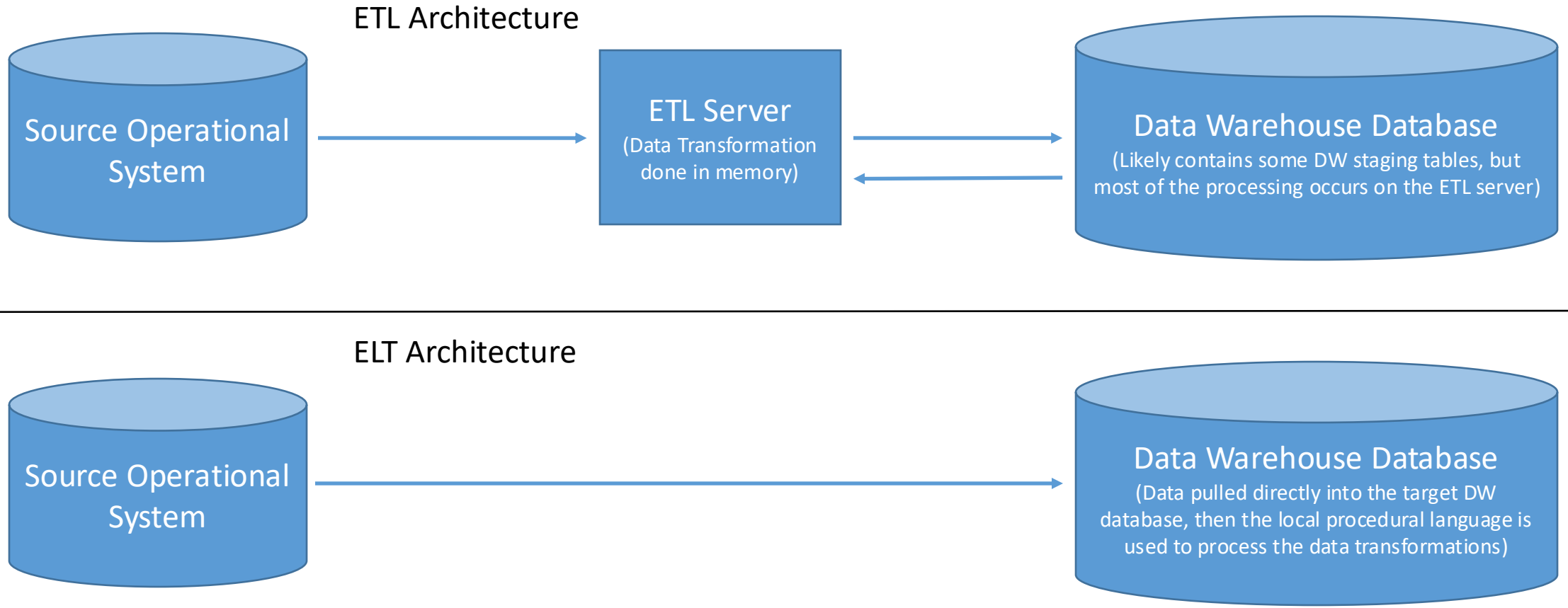
marketed as a Data Automation Tool

- Can be used for rapid development
- Creates Documentation
- Handles ETL type functionality
- Version Control
- Multi-developer aware
- Dependency definition and control
- Process Scheduling

WhereScape is technically an ELT not ETL tool

- ELT tools have a slightly different architecture
- ELT tools handle all of the transformation “heavy processing” on the target DW database versus using a dedicated ETL server and doing the transformations in memory
- ELT architecture leverages the database’s data transformation capabilities versus relying on a proprietary ETL tool to do the transformation work
- Eliminate the need for additional ETL server(s)
- Often perform faster than traditional ETL architectures
- Another popular ELT tool is ODI (Oracle Data Integrator)

ETL versus ELT architecture



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All of the following are true about ELT architecture, except:

A: Relies on the target database engine for transformation processing

B: Does not require large servers (or VMs) dedicated for data transformation processing

C: Is the preferred solution for all target database types

D:Often outperforms traditional ETL architecture for transformation processing

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WhereScape Architecture

WhereScape creates a metadata repository in the target schema of the "DataWarehouse" connection.

- There are 217 repository tables and they are have a prefix of "WS"
- The repository tables contain all of the metadata that defines the Warehouse
  - Table & View column definitions
  - Source to Target attribute mappings
  - Transformation logic
  - Dependency definitions
  - Scheduling information
  - Logging of ETL processing
  - Versioning information

WhereScape connections

- There must be at least one connection defined, and that is the "DataWarehouse" connection.
  - The DW connection is an ODBC connection to the target schema in the DW database.
  - The DW connection is used by WhereScape to "talk to" the metadata repository and handle the target DW ETL functionality
- Additional source connections may be defined, such as
  - Database
  - Hadoop
  - ODBC
  - OS – flat file, XML, Json

WhereScape Objects

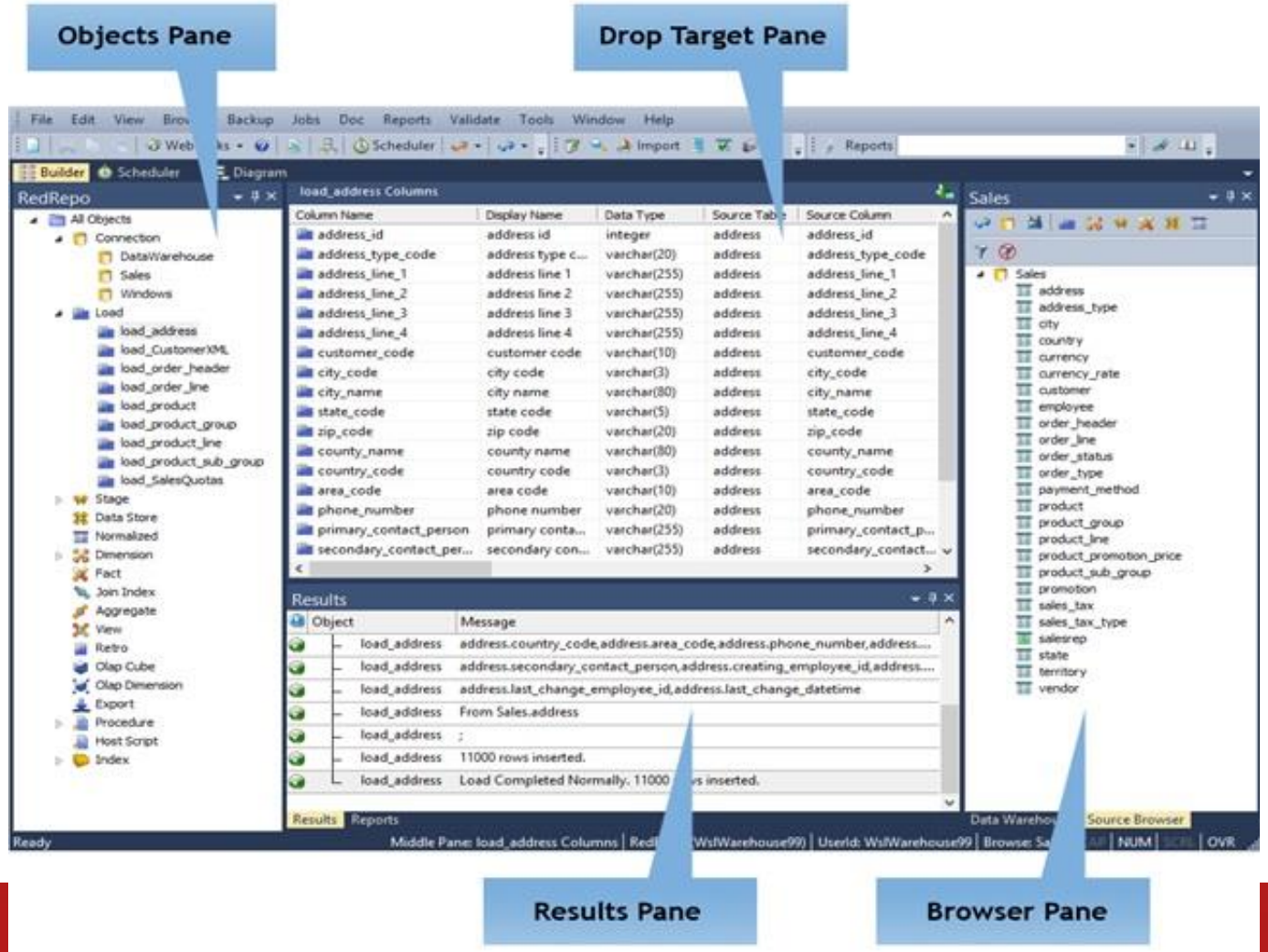
There are 5 object types we will be using in class: Load tables, stage tables, dimension tables, fact tables, and views

- **Load tables** are generally the first step in a WhereScape ETL process.
  - The load table is a copy (with no transformation) from the source system to the warehouse.
  - By simply copying over the source table with no transformation, there is minimal impact on the source system.
  - With a copy of the table in the target Warehouse database, the transformation steps will perform much faster
  - WhereScapes default naming for load tables is to add a "load_" prefix to the table name
- **Stage tables** are the "work horse" of WhereScape ETL processing
  - Stage tables are used for the data transformations
  - Often times there will be multiple stage tables used as part of the ETL processing for a fact or dimension table
  - The default WhereScape naming convention for stage tables is a prefix of: "stage_"
- **Dimension & Fact tables**
  - By default WhereScape assumes incremental processing of these tables from the associated stage tables
  - The default naming convention is a prefix of either "FACT_" or "DIM_"
- **Dimension Views** – Primarily used for Dimension "Role Play"

WhereScape RED Builder Window

- The Builder window/tab has 4 standard panels:
  - The left pane, the **Objects Pane** contains all the objects within the metadata repository
  - Objects include: load tables, stage tables, etc
- The middle pane, the **Drop Target Pane** or **Work Pane,** is used as the drop target in drag and drop operations.
  - The Work Pane is where objects are defined and altered
  - The bottom central Pane shows the results of WhereScape operations
- The right pane, the **Browser Pane**, allows you to browse your **Connections**

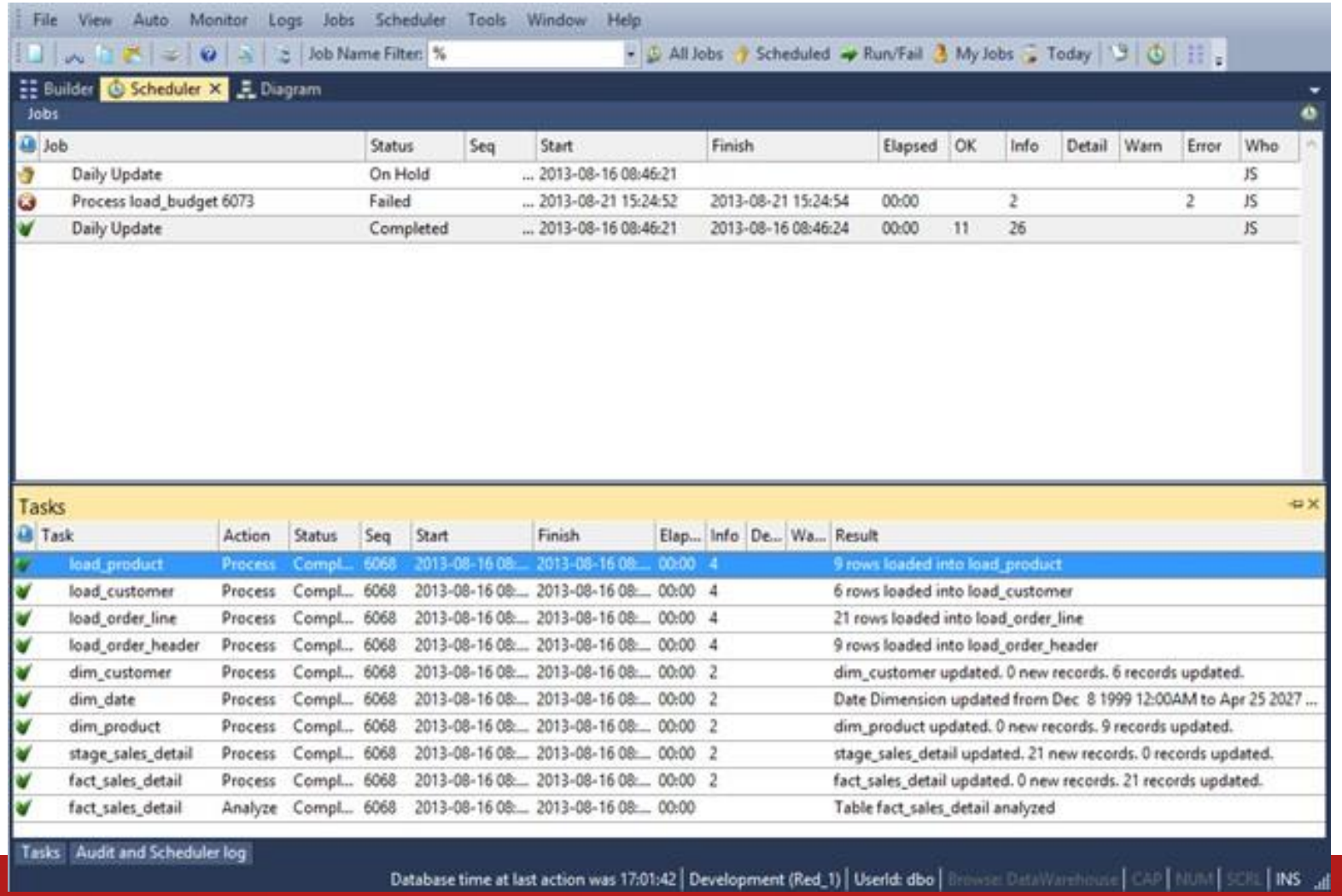
WhereScape RED Builder Window



WhereScape RED Scheduler Window

- Jobs can be scheduled, monitored, edited and deleted through this window.
- The window consists of two panes.
- The toolbar provides a quick way to display various job selections in the top pane.
- Double-click on the job in the top pane to see the tasks of the selected job in the bottom pane.
- Double-click on the task in the bottom pane to see the audit trail displayed in a separate tab in the bottom pane.

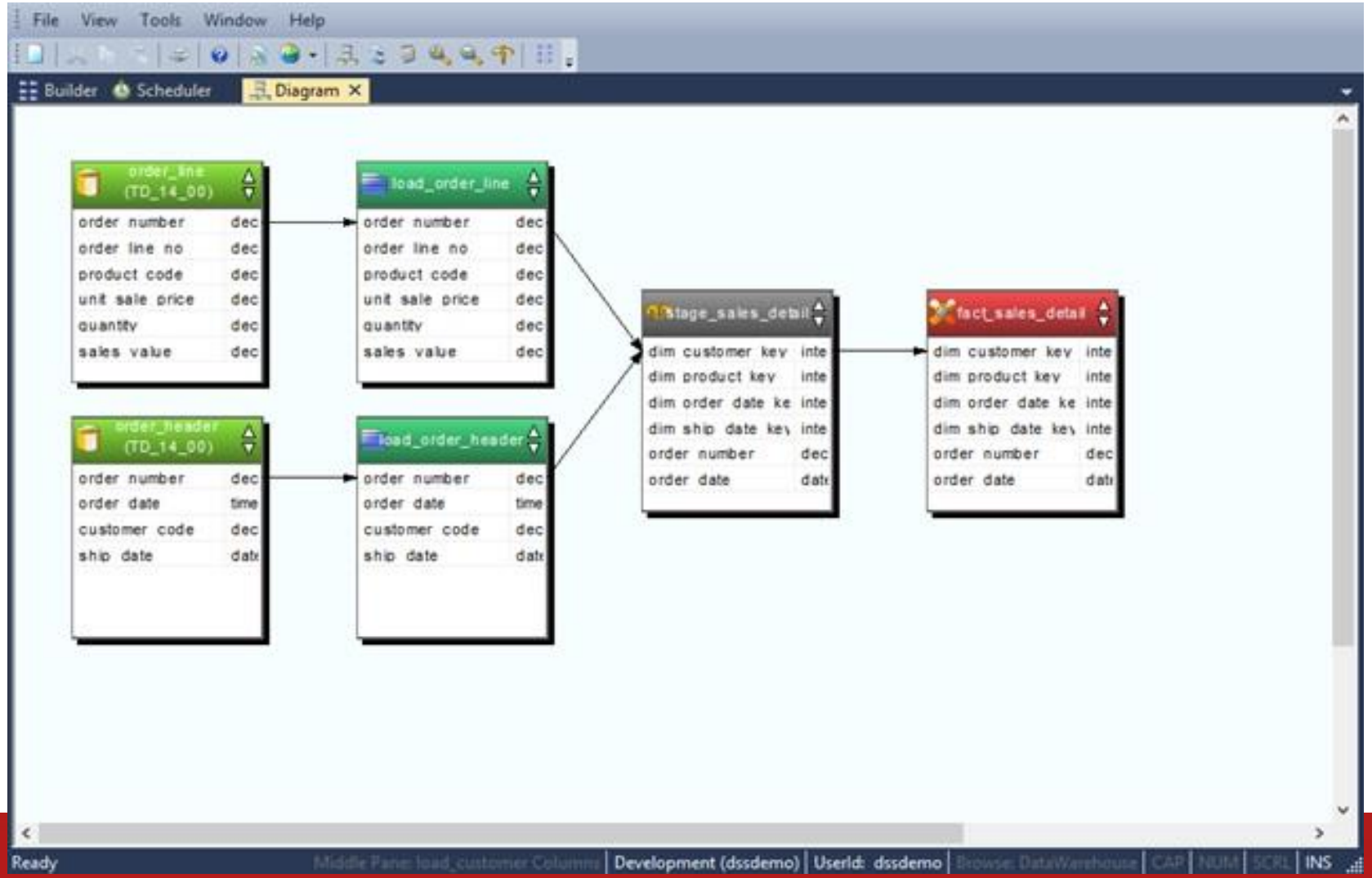
WhereScape RED Scheduler Window



WhereScape RED Diagram Window

- The Diagram Window is used to display the tables of the data warehouse in diagrammatic form, showing the various sources or targets of the selected object.
  - The Diagram Selection is as follows:
    - Schema Diagram (for fact tables, aggregate tables and OLAP cubes)
    - Source Diagram (for any table)
    - Joins Diagram (for any table)
    - Links Diagram (for any table)
    - Impact Diagram (for any table)
    - Dependency Diagram (for any job)

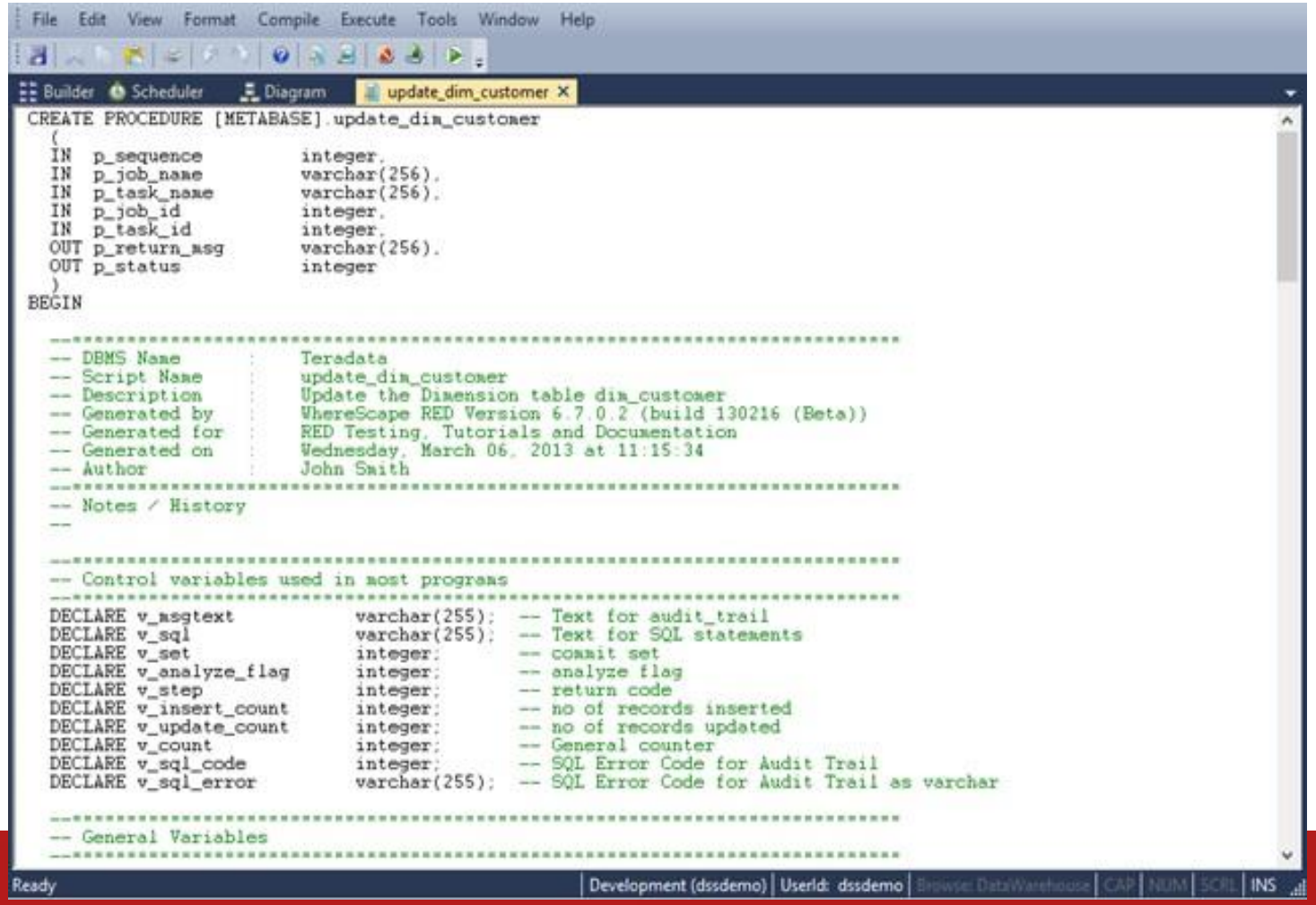
WhereScape RED Diagram Window





WhereScape RED Procedure Editor Window

The Procedure Editor Window provides a means of viewing, editing, compiling, comparing and running procedures





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Review of Assignment 4: WhereScape tutorial



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How are you feeling about the WhereScape RED tutorial / Assignment?

Good – Ready to start

OK – Shouldn't be an issue getting through it by the deadline

Bad – Looks pretty complicated, not sure about this one

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